



Tennessee TECH

Default Question Block

Post-Activity Quiz & Survey After Flagmaker Activity

The questions are the exact same that you answered prior to the activity.

The survey asks you about the flagmaker activity on parallel computing. We will use the results to evaluate these activities for possible use in future terms and at other institutions. The survey should take no more than 8 minutes.

If you have any questions about this survey, please email April Crockett at acrockett@tntech.edu.

Multiple Choice: What is the primary limitation of using only one processor to complete a task?

- It uses too much memory, making the process slower.
 - It can only access local files, making the process slower.
 - It must complete each step one at a time, making the process slower.
 - It causes data corruption.
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Multiple Choice: What is a potential drawback of using multiple processors?

- Multiple processors can cause overheating requiring expensive cooling techniques.
 - Multiple processors can cause the computer to run out of RAM.
 - The processors may finish at different times, causing some to sit idle with no work.
 - The processors are able to complete the task faster.
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True or False: Speedup is defined as the ratio of the time taken to solve a problem on a single processor to the time taken on a parallel system.

True

False

Multiple Choice: What is contention in parallel computing?

- The process of dividing (breaking down) a task into smaller sub-tasks.
 - When multiple processors are waiting on (competing for) shared resources.
 - The increase in computational speed by adding more processors.
 - The ability of a system to handle a growing amount of work.
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True or False: Scalability refers to the ability of a parallel system to increase its performance proportionally with the addition of more processors.

- True
 - False
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Multiple Choice: What is the term for ensuring that each processor has roughly the same amount of work to prevent idle time?

- Multithreading
 - Synchronization
 - Load balancing
 - Memory sharing
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Multiple Choice: What is a race condition in the context of parallel computing?

- Where one processor finishes the task before others start.
 - Where multiple processors try to access or modify shared data simultaneously.
 - A scheduling technique that improves processor speed and synchronization.
 - A timeout that resets processor tasks.
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Multiple Choice: What is pipelining in the context of parallel computing?

- The process of executing multiple tasks simultaneously by multiple processors.
 - The strategy of reducing contention among processors so that multiple processors can't access the same shared resource at the same time.
 - The method of coordinating multiple processors to ensure they work together correctly when accessing shared resources.
 - The technique of dividing a task into sub-tasks and each subtask is handled by a different processor in a sequence like an assembly line.
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To what extent do you agree with each of these statements about the flagmaker activity?

	Strongly Disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I had fun during the activity.	☐	☐	☐	☐	☐
I am confident in my understanding of the material presented during the activity.	☐	☐	☐	☐	☐
The activity increased my understanding of parallel computing.	☐	☐	☐	☐	☐
The activity made me more interested in parallel computing.	☐	☐	☐	☐	☐
The activity increased my understanding of loops.	☐	☐	☐	☐	☐
I was focused during the activity.	☐	☐	☐	☐	☐
I put effort into the activity.	☐	☐	☐	☐	☐

To what extent do you agree with each of these statements related to group engagement?

	Strongly Disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
Explaining the material to my group improved my understanding of it.	☐	☐	☐	☐	☐
Having the material explained to me by my group members improved my understanding of it.	☐	☐	☐	☐	☐
Group discussion during the activity contributed to my understanding of parallel computing.	☐	☐	☐	☐	☐

	Strongly Disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
Overall, the other members of my group made valuable contributions during the activity.	☐	☐	☐	☐	☐
I would prefer to take a class that includes this group activity over one that does not.	☐	☐	☐	☐	☐
I made a valuable contribution to my group during the activity.	☐	☐	☐	☐	☐

To what extent do you agree with each of these statements related to your lab instructors?

	Strongly Disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
The instructor(s) seemed prepared for the activity.	☐	☐	☐	☐	☐
The instructor(s) put a good deal of effort into my learning from the activity.	☐	☐	☐	☐	☐
The instructor(s) enthusiasm made me more interested in the activity.	☐	☐	☐	☐	☐
The instructor(s) were available to answer questions during the activity.	☐	☐	☐	☐	☐

What is the most interesting thing you learned from the activity?

How could we improve the activity for future classes?

Enter your **TTU Username** (example: jdoe42)

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